Quadratics

SAT Math · Foundations to Mastery · Week of 25 October

Name: _____ Due: _____

How to use this sheet

This is four short sittings, not one long one. Four half-hours across the week will do more for you than one long Saturday, and you are far more likely to finish it.

- **This is new ground.** Parts A and B are built so you can finish them. Expect to go slowly in Part C, and expect that to be normal.
- **A squared bracket has two answers, not one.** Whenever you take a square root, write $\pm$ before you write anything else. That single habit is the point of the week.
- When you get one wrong, write it in your error log: what you did, and what you should have done.

Sitting	Part	What it drills	Minutes
Sunday	A	Retention: translation setups	15
Tuesday	B	Expanding and factoring	25
Thursday	C	Solving quadratics	25
Saturday	D, E	In context; six mixed, timed	27

Part A — Retention

No calculator. 15 minutes.

SUNDAY

The first six are last week's work: **write the equation only, do not solve it.** The last four keep weeks two and three alive. Three weeks running at once, in fifteen minutes.

1. A gym charges a \$40 joining fee plus \$29 for each month. Write an equation for the total cost C after m months.

2. Five more than twice a number n is 31. Write the equation.

3. A rectangle's length L is 4 cm less than three times its width W, and its perimeter is 72 cm. Write two equations.

4. A shop sells 150 items in all, hats at \$6 and scarves at \$11, taking \$1,245. Let h be the number of hats and s the number of scarves. Write two equations.

5. A tank holds 400 litres and drains at 12 litres each minute. Write an equation for the volume V left after t minutes.

6. The sum of three consecutive integers is 84. Let n be the smallest. Write the equation.

   _____

7. A line passes through $(-3, 8)$ and $(5, -4)$. Its slope is _____

8. For $4x - 5y = 20$, the y-coordinate of the y-intercept is _____

9. $y = 2x + 1$ and $3x + y = 16$. What is the value of x?
   Answer: _____ *(write your answer)*

10. $5x + 2y = 4$ and $3x - 2y = 12$. What is the value of x?
    Answer: _____ *(write your answer)*

Part B — Expanding and factoring

Calculator allowed. 25 minutes.

TUESDAY

This is the whole of Advanced Math in one skill: moving between $(x + a)(x + b)$ and $x^2 + bx + c$, in both directions. Expanding is multiplication you already know. Factoring is the same thing read backwards — find two numbers that multiply to the last term and add to the middle one. Nothing here is beyond you.

Expand and simplify.

1. $(x + 5)(x + 3)$ _____
2. $(x - 7)(x + 2)$ _____
3. $(2x + 3)(x - 5)$ _____
4. $(x + 6)(x - 6)$ _____
5. $(x - 5)^2$ _____

Factor.

6. $x^2 - 49$ _____
7. $x^2 + 12x + 36$ _____
8. $x^2 + 9x + 20$ _____
9. $x^2 - 3x - 28$ _____
10. $2x^2 - 8x - 24$ _____

Choose one answer.

11. Which expression is equivalent to $3x^2 + 15x - 42$?

    A) $3(x - 7)(x + 2)$
    B) $3(x + 7)(x - 2)$
    C) $3(x + 14)(x - 1)$
    D) $(x + 7)(x - 2)$

12. Which expression is equivalent to $(x-9)^2$?

A) x^2-81
B) $x^2-18x-81$
C) $x^2+18x+81$
D) $x^2-18x+81$

Part C — Solving quadratics

Calculator allowed. 25 minutes.

THURSDAY

A quadratic almost always has *two* answers. Before you start each item, write the number 2 in the margin, and do not move on until you have found both.

Two routes. If one side is already zero, factor and set each bracket to zero. If the equation looks like $(x-h)^2=k$, take the square root of both sides — and write $\pm$ on the right-hand side *as you write it*, not afterwards.

1. $x^2-7x+12=0$. What is the greater of the two solutions?
   Answer: _____ *(write your answer)*

2. $x^2+2x-15=0$. What is the lesser of the two solutions?
   Answer: _____ *(write your answer)*

3. $x^2-5x=24$. What is the greater of the two solutions?
   Answer: _____ *(write your answer)*

4. $x^2=6x+16$. What is the lesser of the two solutions?
   Answer: _____ *(write your answer)*

5. $(x-3)^2=49$. What is the lesser of the two solutions?

   A) -10
   B) -4
   C) 4
   D) 10

6. $(x+2)^2=36$. What is the lesser of the two solutions?

   A) -8
   B) -4
   C) 4
   D) 8

7. $(x+6)^2=25$. What is the lesser of the two solutions?

   A) -11
   B) -1
   C) 1
   D) 11

8. $(x-1)^2 = 64$. What is the lesser of the two solutions?

   A) -9
   B) -7
   C) 7
   D) 9

9. $2x^2 - 7x + 3 = 0$. Use the quadratic formula. What is the greater of the two solutions?

   Answer: _____ *(write your answer)*

10. $3x^2 + 5x - 2 = 0$. Use the quadratic formula. What is the lesser of the two solutions?

    Answer: _____ *(write your answer)*

11. How many real solutions does $x^2 + 4x + 9 = 0$ have?

    A) Zero
    B) Exactly one
    C) Exactly two
    D) Infinitely many

12. What is the sum of the two solutions of $x^2 - 11x + 24 = 0$?

    Answer: _____ *(write your answer)*

Part D — In context

Calculator allowed. 15 minutes.

SATURDAY

Here the algebra gives you two answers and the situation throws one away. A time before the ball was thrown, a width of -10 metres: the equation does not know these are impossible, so you have to. Solve, then look at both answers and ask which one could happen.

1. A ball is thrown from a roof. Its height in feet after t seconds is $h(t) = -16t^2 + 48t + 64$. After how many seconds does it hit the ground?

   Answer: _____ *(write your answer)*

   The other solution is rejected. Why?

   _____

2. A rectangular garden is 3 metres longer than it is wide, and its area is 70 square metres. What is its width, in metres?

   Answer: _____ *(write your answer)*

   The other solution is rejected. Why?

   _____

3. A stall's daily revenue, in dollars, is modelled by $R(p) = -5p^2 + 90p$, where p is the price in dollars. For which lower of the two prices is the revenue exactly \$400?

   Answer: _____ *(write your answer)*

4. A ball is thrown upward and its height in feet after t seconds is $h(t) = -16t^2 + 64t + 5$. What is the greatest height, in feet, that the ball reaches?

   A) 2
   B) 5
   C) 64
   D) 69

5. A stall's daily profit, in dollars, is $P(n) = n^2 + 10n - 1200$, where n is the number of items sold. How many items must it sell to break even, that is, for the profit to be \$0?

   Answer: _____ *(write your answer)*

   The other solution is rejected. Why?

   _____

Part E — Mixed, timed

Calculator allowed. 12 minutes, timed.

SATURDAY

Set a timer for 12 minutes. These span Parts B to D. Answer every one — there is no penalty for a wrong answer, so a blank is a mark thrown away. Where you find two solutions, read the question again before you write one of them down.

1. Which expression is equivalent to $(x-6)(x+2)$?

   A) $x^2 - 12$
   B) $x^2 - 4x - 12$
   C) $x^2 + 4x - 12$
   D) $x^2 - 8x - 12$

2. $x^2 - 9x + 18 = 0$. What is the greater of the two solutions?

   Answer: _____ *(write your answer)*

3. $(x+5)^2 = 81$. What is the lesser of the two solutions?

   A) -14
   B) -4
   C) 4
   D) 14

4. Which expression is equivalent to $4x^2 - 25$?

   A) $(2x-5)(2x+5)$
   B) $(2x-5)^2$
   C) $(4x-5)(x+5)$
   D) $(4x-25)(x+1)$

5. A ball is thrown from a ledge and its height in feet after t seconds is $h(t) = -16t^2 + 32t + 48$. After how many seconds does it hit the ground?

   Answer: _____ *(write your answer)*

6. What is the sum of the two solutions of $2x^2 - 14x + 5 = 0$?

   Answer: _____ *(write your answer)*

Error log

Fill this in as you go.

Date	Q	What I did	What I should have done

Bring this sheet and your error log to the next session.